Causal Inference for Infectious Diseases

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University of North Carolina at Chapel Hill

November 17, 2025

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Chapel Hill
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Charlotte
Greenville
New Bern
Greenville, SC
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CFAR
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STOP

Why Biostatistics?

Humans

Why Biostatistics?

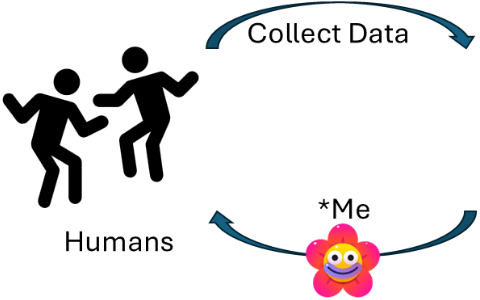
Humans

Collect Data

Data

X	Y	Z
0	100	H
0	150	L
1	100	L
1	150	H

Why Biostatistics?



Data

X	Y	Z
0	100	H
0	150	L
1	100	L
1	150	H

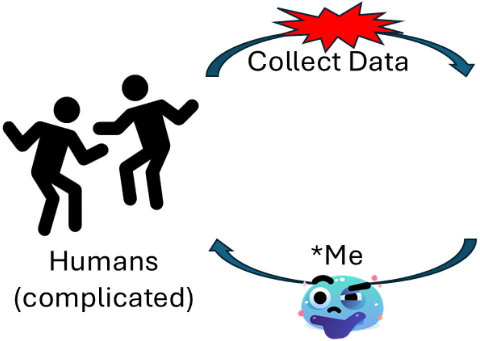
Why Biostatistics?

Humans
(complicated)

Data

X	Y	Z
0	100	H
NA	>100	L or H
NA	100	L or H
1	>100	H

Why Biostatistics?



Data

X	Y	Z
0	100	H
NA	>100	L or H
NA	100	L or H
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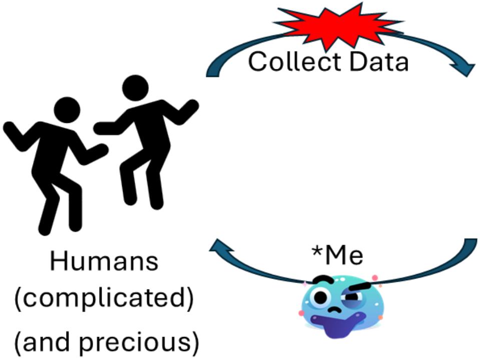

Data

X	Y	Z
0	100	H
NA	>100	L or H
NA	100	L or H
1	>100	H

Example Research Question

Would college students be less likely to spread the flu if they were told to social distance when sick?

eX-FLU Trial

- **eX-FLU**: trial to evaluate a social distancing intervention on a college campus during flu season [1, 2]

Design and methods of a social network isolation study for reducing respiratory infection transmission: The eX-FLU cluster randomized trial

Allison E. Aiello[a,*], Amanda M. Simanek[b,1], Marisa C. Eisenberg[c], Alison R. Walsh[c], Brian Davis[c], Erik Volz[d,1], Caroline Cheng[c], Jeanette J. Rainey[e], Amra Uzicanin[e], Hongjiang Gao[e], Nathaniel Osgood[f], Dylan Knowles[f], Kevin Stanley[f], Kara Tarter[c], Arnold S. Monto[c]

eX-FLU Trial

- **eX-FLU**: trial to evaluate a social distancing intervention on a college campus during flu season [1, 2]
- **Intervention**: encouragement to isolate in dorm for three days upon developing symptoms of **influenza-like illness** (ILI)

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eX-FLU Trial

- **eX-FLU**: trial to evaluate a social distancing intervention on a college campus during flu season [1, 2]
- **Intervention**: encouragement to isolate in dorm for three days upon developing symptoms of **influenza-like illness** (ILI)
- **Central question**: does the encouragement-to-isolate intervention reduce transmission of ILI?

Design and methods of a social network isolation study for reducing respiratory infection transmission: The eX-FLU cluster randomized trial

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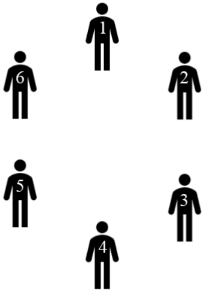
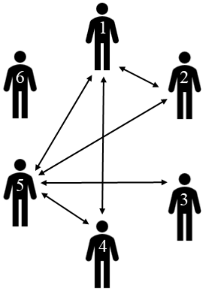
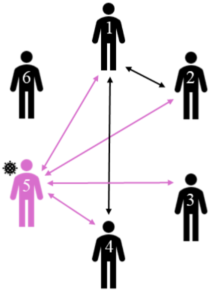
Transmission of Influenza-Like-Illness



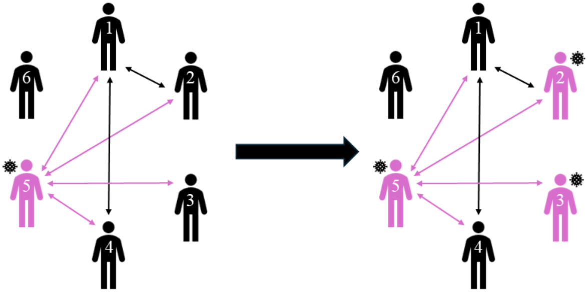
Transmission of Influenza-Like-Illness



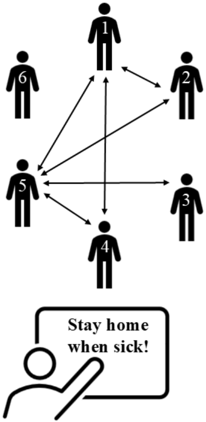
Transmission of Influenza-Like-Illness



Transmission of Influenza-Like-Illness



Example I: Intervention Affects Network and Transmission



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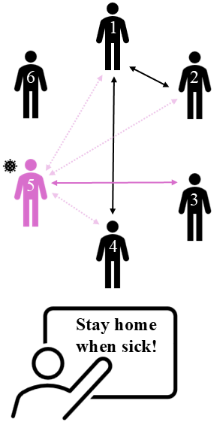

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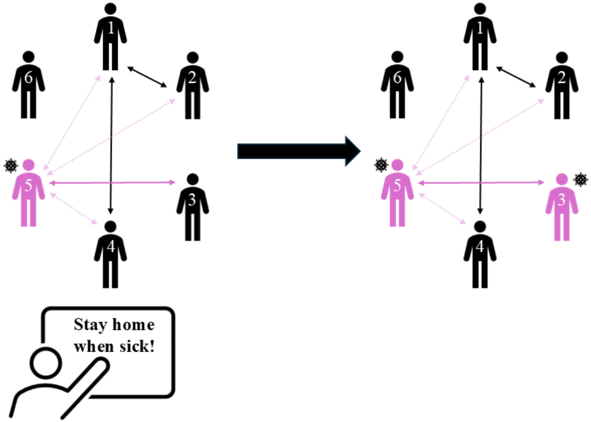
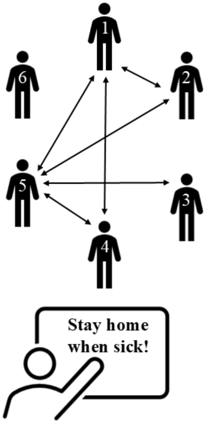


Example II: Intervention Affects Network, Not Transmission



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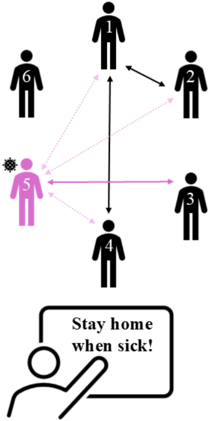

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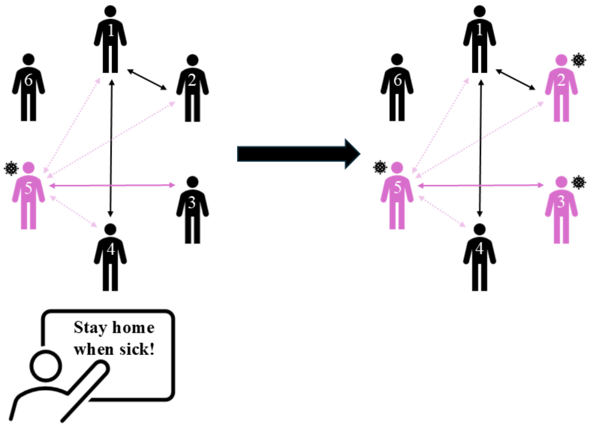
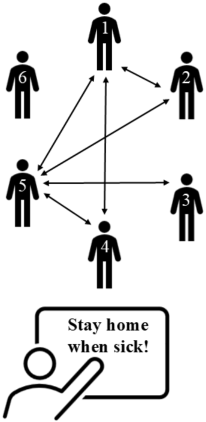


Example III: Intervention Affects Transmission, Not Network



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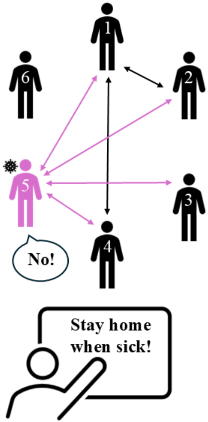
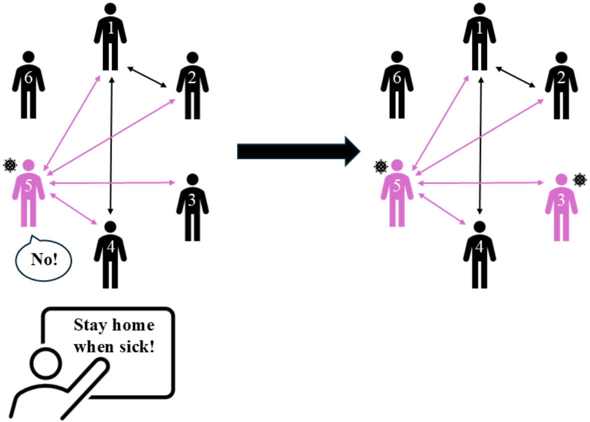
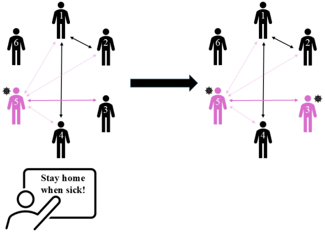


Example III: Intervention Affects Transmission, Not Network

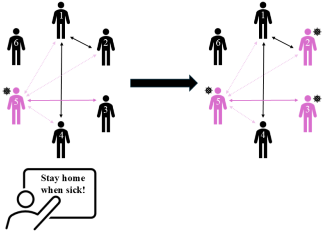


Does the Intervention Affect Transmission of ILI?

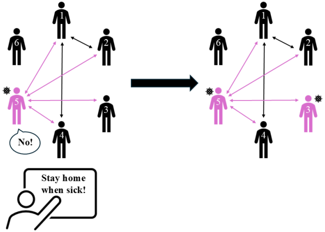
Example I:



Example II:

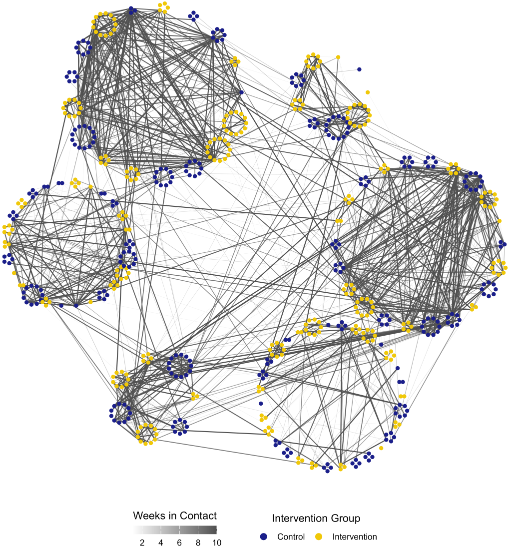


Example III:

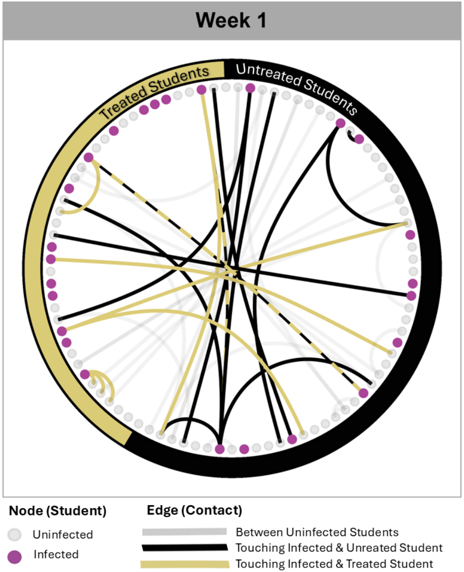


In Examples I and III, the answer is yes

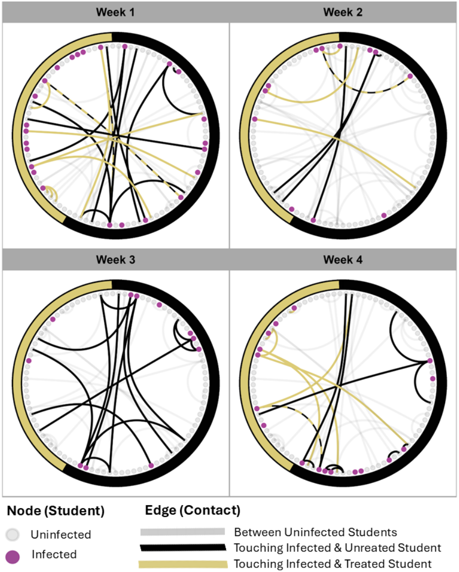
eX-FLU Data


Weeks in Contact
2 4 6 8 10
Intervention Group
Control Intervention

eX-FLU Data


Week 1
Treated Students
Untreated Students
Node (Student)
Uninfected
Infected
Edge (Contact)
Between Uninfected Students
Touching Infected & Unreated Student
Touching Infected & Treated Student

eX-FLU Data


Week 1
Week 2
Week 3
Week 4
Node (Student)
Uninfected
Infected
Edge (Contact)
Between Uninfected Students
Touching Infected & Unreated Student
Touching Infected & Treated Student

References

[1] Allison E. Aiello, Amanda M. Simanek, Marisa C. Eisenberg, Alison R. Walsh, Brian Davis, Erik Volz, Caroline Cheng, Jeanette J. Rainey, Amra Uzicanin, Hongjiang Gao, Nathaniel Osgood, Dylan Knowles, Kevin Stanley, Kara Tarter, and Arnold S. Monto. Design and methods of a social network isolation study for reducing respiratory infection transmission: The eX-FLU cluster randomized trial. *Epidemics*, 15:38–55, June 2016. ISSN 17554365. doi: 10.1016/j.epidem.2016.01.001. URL https://linkinghub.elsevier.com/retrieve/pii/S1755436516000025.

[2] P.N. Zivich, M.C. Eisenberg, A.S. Monto, A. Uzicanin, R. S. Baric, T. P. Sheahan, J. J. Rainey, H. Gao, and A. E. Aiello. Transmission of viral pathogens in a social network of university students: the eX-FLU study. *Epidemiology and Infection*, 148:e267, 2020. ISSN 0950-2688, 1469-4409. doi: 10.1017/S0950268820001806. URL https://www.cambridge.org/core/product/identifier/S0950268820001806/type/journal_article.